

– STARTING SHORTLY –

Why Agents? The case for *autonomous AI.*

In the next two hours you'll name the five gaps in your chatbot, run a four-question funnel against any business problem, and build your first agent — no code — that does what a chatbot cannot.

– IDENTIFY GAPS – CLASSIFY PATTERNS – QUANTIFY ROI – BUILD AN AGENT

— COLD OPEN · 2 MIN

In one word— *how did Session 1 land for you?*

Before we say anything else. One word in the chat. Pick the one closest to honest — the facilitator reads every reply by name.

Useful

Overwhelming

Sticky

Surprising

Skeptical

Curious

Other → type it

NO WRONG ANSWERS. PICK THE CHIP CLOSEST TO TRUE, OR TYPE YOUR OWN.

— THE BAKERY THAT FAILED · THE GAP WE CLOSE

4 of 6 steps it couldn't do. *Today you build the thing that doesn't.*



...if you remember the bakery campaign from Session 1. (The one ChatGPT failed on 4 of 6 steps.)

— WHAT YOU WALK OUT WITH TODAY —

1. The **five gaps** every chatbot has — named, numbered, severity-ranked.
2. The **four-question funnel** for Chatbot · RAG · Agent · Multi-Agent.
3. The **real business cost** of staying on chatbots — three failure scenarios.
4. A working **AI Pattern Advisor** agent on your screen — yours, role-customized, no code.

— SESSION 2 OF 8 · THE CASE FOR AUTONOMOUS AI

— WHY IT MATTERS NOW —

COMPANIES ABANDONING AI PROJECTS

42%

in 2025 — up from 17% in 2024

SOURCE · [FULLVIEW AI STATISTICS 2025](#)

ACHIEVING FIRST-YEAR ROI ON AI

74%

when the pattern match is right

SOURCE · [MCKINSEY STATE OF AI 2025](#)

Every failure has a fix. That fix is called an agent — when you reach for the right pattern.

— SESSION 1, FINAL SLIDE

— HOUSE RULES + THE PATH THROUGH

Four norms. *Four moves.*

— CAMERAS —

Encouraged, optional

No one called out for off-camera.

— CHAT —

Use it freely

Questions, reactions, jokes — welcome.

— MICS —

Muted unless called

Or unmute to share. Either works.

— BREAK —

Real 10 min, ~halfway

Cameras off. Stretch. Water. Non-negotiable.

— STUCK QUESTIONS —

Ask live

Longer answers come by email after class.

— NEW FOR TODAY —

More Builder than slide

You spend more time in EdYoda Agent Builder than watching the screen. Bring your hands.

Concepts

→

Build

→

Debrief

→

Done

TWENTY MINUTES OF CONCEPTS. SIXTY-FIVE MINUTES OF BUILD. TEN MINUTES OF DEBRIEF. THEN CLOSE. THE THING YOU BUILD IS CALLED THE **AI PATTERN ADVISOR** — SAME SHAPE FOR EVERYONE, ROLE-CUSTOMIZED FOR MARKETING/SALES · FINANCE/CONSULTING · DOCTOR/HEALTHCARE · GENERIC.

— ONE RULE FOR TODAY —

You will not leave with *notes*.
You will leave with a *working*
agent.

SIX SPRINTS NEXT. TWENTY MINUTES. RECOGNIZE, NOT MEMORIZE.

The five gaps *agents close*.

*What chatbots cannot do — and how autonomous AI agents fix each limitation.
Ordered by severity. Numbered in the order we'll close them as a class today.*

#1	Memory gap	Chatbot forgets everything after the session ends. Agent persists context across runs, hours, days.	CRITICAL
#2	Multi-step reasoning	Chatbot handles one instruction at a time. Agent manages chains of dependent decisions.	CRITICAL
#3	Tools gap	Chatbot only produces text. Agent calls APIs, reads files, writes to real systems.	HIGH
#4	Autonomy gap	Chatbot waits for input on every step. Agent continues until the goal is complete.	HIGH
#5	Planning gap	Chatbot answers one question. Agent decomposes goals into ordered steps with a time horizon.	MEDIUM

Each gap is a class of work your chatbot silently refuses to do. Naming them is half the fix — closing them is the rest of today. Watch for **Gap #3** in Step A, **Gap #4** in Step C, **Gap #1** in Step D, and **Gaps #2 + #5** in Step E.

— SPRINT 02 · THE MECHANIC

From *prompt* → *response* to *goal* → *plan* → *act* → *observe* → *reflect* → *iterate*.



CHATBOT · THE VENDING MACHINE

You press a button, you get one thing. No follow-through, no second move.

AGENT · THE PROJECT MANAGER

You give a goal, they plan, execute, check the result, adjust, finish.

The agent maturity model — *Level 0 to Level 3.*

Progressive evolution from basic chatbots to coordinated multi-agent systems.

L0	Chatbot Single turn. No memory. No tools. The interface most people equate with "AI". Example · FAQ bot on a website	25%
L1	Copilot Suggests next steps inline. Human reviews and approves every action. Example · GitHub Copilot · sales email assistant	50%
L2	Agent <u>TODAY'S BUILD</u> Plans, uses tools, persists memory, completes multi-step goals on its own. Example · AI Pattern Advisor · recruitment screener	75%
L3	Multi-Agent An orchestrator coordinates specialised agents working in parallel. Example · research pipeline · scout → analyst → writer	100%

Most enterprises sit at *L0-L1*. The competitive advantage is in moving to *L2* — which is exactly where you'll be by the end of this session.

— SPRINT 04 · THE DISCIPLINE

The four-question funnel — *chatbot, RAG, agent, multi-agent.*

— ASK IN ORDER. ONE SURVIVES. —

- 1

Does the answer exist in a fixed document or database?

YES → RAG

NO → CONTINUE
- 2

Is one response — with no follow-up action — enough?

YES → CHATBOT

NO → CONTINUE
- 3

Does it require multiple steps, memory, or tool use?

YES → AGENT

NO → CONTINUE
- 4

Does it need multiple specialised capabilities in parallel?

YES → MULTI-AGENT

— THE FOUR SHAPES —

RAG	Document retrieval. Grounded answers from a fixed corpus.
Chatbot	Single response. One question, one reply, no follow-through.
Agent	Multi-step tasks. Plans, uses tools, persists memory.
Multi-Agent	Parallel execution. Specialists orchestrated toward one goal.

Three questions eliminate. Only one path survives. Most teams skip this discipline and overbuild — **95%** pattern-matching success when you don't.

TRY IT — classify each one

1.

Answer customer questions about return policy.

2.

Generate a one-time product description.

3.

Screen 200 resumes and schedule interviews.

4.

Run a competitive intelligence report.

R = RAG · C = Chatbot · A = Agent · M = Multi-Agent

— SPRINT 05 · THE PRICE

The cost of *staying* on chatbots.

Three failure shapes you'll recognise from your own work — and the survey data on what they're costing the market.

— MISSED CONTEXT —

Support that forgets

CUSTOMER SUPPORT

Agent gives wrong advice because it has no memory of prior interactions. Tickets escalate. CSAT drops.

— NO FOLLOW-THROUGH —

Deals that stall

SALES PROCESS

Chatbot drafts a proposal but cannot send it, cannot track it, cannot schedule the follow-up. Cycles extend, deals go cold.

— ZERO PERSISTENCE —

Workflows that reset

OPERATIONS & COMPLIANCE

Session times out, all context is lost mid-workflow. The work that just got done has to be redone — by hand.

COMPANIES ABANDONING AI PROJECTS

42%

in 2025 — up from 17% in 2024 (MIT / RAND, 2025)

SOURCE · [FULLVIEW AI STATISTICS 2025](#)

ACHIEVING FIRST-YEAR ROI ON AI

74%

of organisations report ROI in year one

SOURCE · [MCKINSEY STATE OF AI 2025](#)

EXPERIMENTING WITH AI AGENTS

62%

of organisations — high-performers are 3× more likely to be scaling

SOURCE · [MCKINSEY STATE OF AI 2025](#)

10^{/28}

Five agentic *architecture patterns*.

Five proven shapes for building autonomous AI systems — and when to use each.

Single Agent

TODAY'S BUILD

— SELF-CONTAINED —

One LLM + tools + memory for sequential tasks.

When the task is self-contained and sequential.

→ Email drafting agent

Tool-Calling Agent

API

— API INTEGRATION —

Agent with external API integrations.

When the task reads or writes external systems.

→ CRM data sync

Human-in-the-Loop

HIGH-STAKES

— CHECKPOINT APPROVAL —

Agent pauses at checkpoints for human approval.

When the task is high-stakes or regulated.

→ Financial approvals

RAG Agent

DOCS

— DOCUMENT-BASED —

Single agent with knowledge-base retrieval.

When the task requires grounding in documents.

→ Support FAQ bot

Orchestrator-Worker

MULTI

— MULTI-AGENT —

One planning agent coordinates specialists.

When the task has parallel workstreams.

→ Market research pipeline

Up next

BRIDGE

— BUILD COMES NEXT —

Concepts named. Time to build the Single-Agent + Tool-Calling pattern — the one underneath every other shape.

Block 2 begins now.

→ Your AI Pattern Advisor

— BUILDER ORIENTATION

Four places to know. *One button.*

The EdYoda Agent Builder. Don't open it yet — just learn the shape so you know where everything lives when we start.

— TAB 01 —

Overview

Agent identity — name, description (role), welcome message.

What it IS · written once

— TAB 03 —

RAG Docs

Upload docs the agent retrieves from. *This IS the memory.*

Persistent context · one upload

— TAB 02 —

Configure → No-Code

The pipeline of nodes. This is where most of the build lives.

What it DOES · 5 nodes today

— TAB 04 —

Logs

Run history. Look here after you publish — not during the build.

Observe · later, not now

— TRACK SELECTION

Same agent shape. *Different words.*

*You're building the **AI Pattern Advisor** — an agent that classifies any business problem into the right AI pattern. Pick the track closest to your day job.*

01

Marketing/Sales

AI Pattern Advisor for marketing & sales workflows.

02

Finance/Consulting

AI Pattern Advisor for financial workflows.

03

Doctor/Healthcare

AI Pattern Advisor for clinical workflows.

04

Generic

AI Pattern Advisor for general knowledge-work.

13/28

A BUILD STEP · HANDS-ON
Ten minutes

Create the agent + *Node 1 (Web Search Tool Agent).*

- 01 On the Agents list, click + **New**. Fill the modal:
- 02 Agent Name: [YourName]'s AI Pattern Advisor - [Track]

03 Description: paste the **role line** from your workbook page (4–7)

04 Welcome Message: paste the **welcome line** from your workbook page

05 Framework: anthropic
- 06 Click **Create Agent → Confirm & Create**.
- 07 On the agent page, open **Configure → No-Code**.
- 08 Configure Node 1: Name = Web Search Tool Agent · Model = Claude Sonnet 4.6 · Max Tokens = 1024 · System Prompt = paste Node 1's prompt from your workbook page.
- 09 Stop here. **Do not click Proceed yet** — we run Step B before adding more nodes.

*One node, one job — research. After this step your agent can use web search, which the chatbot in Sprint 1 couldn't. **Gap #3 Tools — closed.***

B

BUILD STEP · HANDS-ON

Six minutes

First run — *proof of life.*

- 01 Configure → No-Code → click **Proceed (Generate Code)**.
- 02 Click **Publish Agent** (top right).
- 03 Copy the **public chat URL** from Share Config. Open it in a new tab.
- 04 Paste your track's **starter problem** (workbook page 4–7) into the chat.
- 05 First response may take **15–60 seconds** on a cold start — wait, don't refresh.
- 06 Watch Node 1 fire — web search → structured research output (problem restated, current practice, tools, sources).
- 07 Paste the **HEADLINE INSIGHT line** from your Node 1 output in chat.

15/28

*Just one node so far. Research only — no classification yet. That's Node 2's job, after the break. **Don't worry if the output isn't a pattern recommendation — it isn't supposed to be.***

- B R E A K -

10:00

Stand up. Water. Stretch.

- MEMORY, ACTION, MULTI-STEP. SEE YOU ON THE OTHER SIDE.

16^{/28}

— PROGRESS TRACKER

Three more steps to *L2*.

You walked back from the break with a running L1. Now we upgrade. Keep this slide in the corner of your eye — we move through it together.



...if you're back at your desk. We start Step C once most of the room is back.



DONE · BEFORE THE BREAK

Node 1 (Web Search Tool Agent) · published & running

COMPLETE



NEXT · STEP C

Add Node 2 — the Pattern Classifier

IN PROGRESS



THEN · STEP D

Upload a RAG Doc with your real context

PENDING



THEN · STEP E

Add Nodes 3, 4, 5 · Proceed · Republish

PENDING

WHEN ALL FOUR ARE CHECKED, YOUR PUBLISHED 5-NODE PIPELINE IS LIVE.
THAT'S L2 — THE LINE YOU CAME TO CROSS.

C

BUILD STEP · HANDS-ON

Eight minutes

Add Node 2 — *the Pattern Classifier.*

- 01 Configure → No-Code → click + **Add Node**.
- 02 Configure Node 2: Name = · Model = Claude Sonnet 4.6 · Max Tokens = 1024 · System Prompt = paste Node 2 from your workbook page (4–7).
- 03 Click **Proceed (Generate Code)**.
- 04 Open your published chat URL. Re-run the same starter problem from Step B.
- 05 Paste in chat: the **PATTERN** your agent classified (RAG / Chatbot / Agent / Multi-Agent).

18/28
*Node 1 researches. Node 2 decides — applying the 4-question funnel to Node 1's output. Two nodes, one button-press, no human in the middle. **Gap #4 Autonomy** — **closed.***

D

BUILD STEP · HANDS-ON
Seven minutes

Upload a RAG Doc — *give the agent your real context.*

- 01 Open the **RAG Docs** tab on your agent.
- 02 Use the template on **workbook page 8**. Fill it with your real company, audience, stack, top-3 priorities.
- 03 Save as or . Click upload.
- 04 Re-run the SAME starter problem on your published chat URL.
- 05 Paste in chat: **ONE specific phrase** from the new output that proves the agent used your RAG doc.

19/28

*On this platform there is no memory toggle. **The doc IS the memory.** Upload once, the agent retrieves from it on every future run. **Gap #1 Memory — closed.***

E

BUILD STEP · CLIMAX
Ten minutes

Add Nodes 3, 4, 5 — *publish the pipeline.*

01 Configure → No-Code → click + **Add Node** three times.

02 From your workbook page (4–7), paste in order:

- 03 **Node 3** — Implementation Sketch
- 04 **Node 4** — Risk & Compliance
- 05 **Node 5** — Executive Brief

06 Click **Proceed (Generate Code)**, then click **Republish**.

07 Open your published chat URL. Paste your starter problem. Watch **all 5 nodes** fire in order.

08 Paste in chat: the **NEXT STEP FOR THE FOUNDER** line from your Node 5 output.

*Five nodes. One prompt. One published URL. The agent researches, decides, plans, flags risks, composes a brief — all on its own. **Gaps #2 Multi-step + #5 Planning — closed. Five for five.***

— SENSE-MAKING

You built it. *Now let's name what you built.*

Four lenses from the start of class. Map your agent onto each — this is how you'll explain it to your CTO on Monday.

— LENS 01 —

5 Gaps Scorecard

— LENS 02 —

Maturity Ladder

— LENS 03 —

4-Question Funnel

— LENS 04 —

Pattern You Used

SAME FOUR FRAMEWORKS YOU SAW ON THE WAY IN. NOW RUN THEM BACKWARDS
THROUGH WHAT YOU BUILT.

— LENS 01 · 5 GAPS SCORECARD

Five for five. *In 90 minutes.*

Remember the five chatbot gaps from slide 6? Walk through what your agent now does for each one.

✓	#1 Memory	You enabled persistent memory in Step D — the agent remembers across runs.	CLOSED
✓	#2 Multi-step	Step E proved it — the trace log shows each step referencing the last.	CLOSED
✓	#3 Tools	You added two — Web Search in Step A, an action tool in Step C.	CLOSED
✓	#4 Autonomy	From Step C onwards it acted without per-step approval — you handed it the goal, it ran.	CLOSED
✓	#5 Planning	Step E forced a three-step-plus goal — the agent decomposed the work itself.	CLOSED

5/5

All five gaps closed in ninety minutes — no code, one builder, a working agent. That is the bar.

22/28

L2. You are here.

The maturity ladder from earlier today. Your agent plans, uses tools, persists memory, completes multi-step goals on its own — that's Level 2 by definition.

L0	Chatbot Single turn. No memory. No tools. ✓ Passed — you left this behind on slide 6.	25%
L1	Copilot Suggests next steps. Human approves every action. ✓ Passed — this was your state at the break.	50%
L2	Agent ✓ <u>YOU ARE HERE</u> Plans, uses tools, persists memory, completes multi-step goals on its own. → Your AI Pattern Advisor lives here.	75%
L3	Multi-Agent An orchestrator coordinates specialised agents in parallel. Not today — you'd add this if one agent isn't enough.	100%

L2 SOLVES PROBABLY 80% OF THE WORKFLOWS YOU'D WANT TO AUTOMATE AT WORK.
DON'T OVER-ENGINEER.

Through the funnel: *Agent*.

Take your AI Pattern Advisor through the four-question funnel from slide 9. Watch where you land.

1 Does the answer exist in a fixed document or database?

YES → RAG

NO → CONTINUE ↓

2 Is one response — no follow-up — enough?

YES → CHATBOT

NO → CONTINUE ↓

3 Does it require multiple steps, memory, or tool use?

YES → AGENT ✓

→ You stop here. This is your agent.

4 Does it need multiple specialised capabilities in parallel?

NOT REACHED — YOU EXITED AT Q3

— WHERE YOU LANDED —

RAG

You don't have a fixed corpus — the problem space is open.

Chatbot

A single reply wouldn't solve a multi-step classification.

Agent ✓

Your AI Pattern Advisor. Plans, uses tools, persists context.

Multi-Agent

Save it for when one agent isn't enough — not today.

The funnel killed two options and survived one. That is the discipline. Use it before saying "let's use AI for X" at work.

Single Agent + Tool-Calling. The foundation pattern.

Two patterns from slide 11, fused. Everything else is an upgrade you bolt on when you hit a wall — not before.

Single Agent

YOU BUILT THIS ✓

— SELF-CONTAINED —

One LLM + tools + memory for sequential tasks.

→ *Foundation pattern. The base every upgrade builds on.*

→ Your AI Pattern Advisor

Reflection

UPGRADE · ADD

— SELF-REVIEW —

Agent reviews its own output before returning it.

When you hit the same wall twice — add this.

→ Quality check loop

Orchestrator-Worker

UPGRADE · ADD

— MULTI-AGENT —

A planning agent coordinates specialist agents.

When parallel workstreams emerge — split this.

→ Research → Analysis → Write

Tool-Calling

YOU BUILT THIS ✓

— API INTEGRATION —

Agent with external API integrations — the "hands" you added.

→ *Foundation pattern. What made your agent act.*

→ Your two tools

RAG Agent

UPGRADE · ADD

— DOCUMENT GROUNDING —

Add a knowledge base; ground answers in documents.

When you need to ground in a fixed corpus — add this.

→ Policy / SOP retrieval

Human-in-the-Loop

UPGRADE · ADD

— APPROVAL GATE —

Agent pauses at checkpoints for human approval.

When the action is high-stakes — add a gate.

→ Financial / clinical sign-off

— SHOW & TELL

Three of you. *Two minutes each.*

Show your agent run live. Name. Role. What it does. Run. Narrate the trace log. That is the whole thing.

01

— VOLUNTEER 01 —

Name · Track · What it does · Live run · Trace narration

🕒 2:00 max

02

— VOLUNTEER 02 —

Name · Track · What it does · Live run · Trace narration

🕒 2:00 max

03

— VOLUNTEER 03 —

Name · Track · What it does · Live run · Trace narration

🕒 2:00 max

— GO DO THIS

Three moves *this month.*

The build is done. The habit is what matters now.

01

THIS WEEK

Run your agent **five times** on real business problems. Track time saved (workbook page 11).

02

NEXT TWO WEEKS

Iterate the system prompt. Treat it like a **junior employee you're coaching** — specific feedback, not vibes.

03

NEXT MONTH

Show it to one colleague. Find **ONE** workflow they'd want classified. **Build the second.**

→ NEXT · SESSION 03

Prompt Engineering @ Context Engineering for Agents. Learn how to communicate with agents precisely enough to make them reliable — the discipline that turns today's working agent into a trustworthy one.

TAKE WITH YOU · reply by Friday

1. **ONE business problem at your job** you'll classify with your AI Pattern Advisor this month. One sentence on the problem, one on the pattern you think it'll come back as.
2. **ONE workflow you WON'T automate with an agent** — because it's a chatbot job, a RAG job, or too sensitive for an agent today. One sentence on the workflow, one on why it stays manual.

— — WHERE WE ARE IN THE ARC —

Last session, a chatbot was a *brain in a jar*.

Today, you gave it *hands*.

Next time, we teach it to *listen*.

Two down. Six more to go. The arc gets sharper from here.

Go run your agent on something real this week. That's the only homework that matters.

— SEE YOU IN SESSION 03 · PROMPT ENGINEERING & CONTEXT ENGINEERING FOR AGENTS.